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Identifiability-Aware Virtual Force Sensing and Self-Calibration for Quadrotors

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ABSTRACT Many aerial tasks, such as flying in wind or carrying a payload, require the external force on the airframe, yet quadrotors rarely carry a force sensor. We turn the onboard inertial measurement unit into a *virtual force sensor*: the external specific force is read directly from the accelerometer residual. The obstacle is calibration, since the reading needs the vehicle mass, which is unidentifiable near hover, where a mass error is indistinguishable from a vertical force; a standard extended Kalman filter then corrupts the mass. We make the sensor *identifiability-aware and self-calibrating*. An online identifiability measure, the Fisher information on the mass, gates the calibration and freezes the mass when it cannot be seen; and when the flight lacks the excitation to recalibrate, the controller injects a minimal probe that the same measure shapes. In simulation the sensor reads the external force with a per-axis correlation of 0.93 under continuous wind; the gate holds the mass calibration to $3\,\%$ through a windy hover where an ungated filter drifts to $7\,\%$; the probe recovers the mass from a wrong prior that hover alone cannot identify; and feeding the sensed force to a model-predictive controller reduces the tracking error by 34 to $80\,\%$ under wind, most where the wind dominates the dynamics. The sensor degrades gracefully under injected accelerometer and thrust errors and runs in real time at $100\,$Hz.

INDEX TERMS Aerial robotics, force estimation, inertial sensing, identifiability, self-calibration, virtual sensors.

I. INTRODUCTION

THE external force acting on a quadrotor, from wind, a payload, or physical contact [1]–[3], governs how it must be flown, yet small aerial platforms rarely carry a force or wrench sensor: the payload and power budget favor reusing the sensors already on board. The accelerometer offers a route. It measures the non-gravitational specific force directly, so once gravity and the commanded thrust are removed, what remains is the external specific force. This turns the inertial measurement unit (IMU) into a *virtual force sensor*, with no added hardware, and the reading is instantaneous: no double-differentiation of position, no estimator lag.

The virtual sensor has one calibration parameter, the vehicle mass, which converts the measured acceleration into the thrust contribution that must be subtracted. The mass is rarely measured online, it is copied from a datasheet or changed by a payload, and a few-percent error biases the reading. Worse, the mass is only *conditionally identifiable*: the mass enters the reading only through the thrust-to-weight ratio, so in near-hover flight a mass error is indistinguishable from a vertical external force, and an estimator that calibrates it unconditionally corrupts the very reading it supports. A virtual force sensor for aerial use therefore needs to know *when* its calibration is reliable and to be able to recalibrate itself when it is not. That is the problem we address.

Online parameter identification for aerial robots has been studied with maximum-likelihood frameworks [4] and with recursive regression on inertial signals that recover the mass and inertia under physical consistency constraints [5], [6]; these methods need excited flight to identify the inertial parameters. External-force estimation has a parallel line of work based on momentum observers [7]–[10], which estimate the force on the airframe from the dynamics but are biased by errors in the same parameters above. Moving-horizon estimation has been applied to quadrotor system identification [11], building on the constrained-MHE theory [12]; the unmodeled aerodynamics that learned residual models capture are themselves an active area [13], [14]. On the control side, adaptive and incremental schemes reject disturbances without an explicit force estimate [15]–[19], and learning-based adaptive controllers fit a disturbance representation online to reject wind in agile flight [20]. These methods cancel the disturbance but do not reason about *when* the underlying parameters are identifiable.

Two themes run through this work. First, the mass is recoverable only under sufficient excitation, a persistency-of-excitation condition that classical optimal input design addresses by shaping the input to maximize the information about the unknown [21]. Freezing a parameter update when it is not persistently excited is itself classical in adaptive estimation, in the form of dead-zone and covariance-management schemes [22], [23], and exciting the input to restore information is classical optimal input design [21]. Our contribution is not either element alone but the signal that drives both: unlike a dead-zone keyed to a generic excitation proxy, and unlike a momentum observer that estimates the force without quantifying its identifiability, we gate on the closed-form Schur complement, which (i) isolates exactly the mass–vertical-force coupling, (ii) is the marginal Fisher information on the mass (its inverse is the Cramér–Rao bound), and (iii), in its dimensionless form $\tilde{\sigma}$, doubles as the controller's trigger to restore identifiability. Table 1 makes the comparison explicit: for each family of methods it lists what is measured, which signal (if any) decides whether the parameter is updated, whether the method isolates the mass–vertical-force coupling, and whether it acts to restore identifiability when the flight does not provide it. Extended-state and disturbance observers [17], [24] estimate a lumped disturbance with the model, and hence the mass, assumed known: their disturbance state is the same residual our force reading is built on, so what we add is not a different observer but the calibration layer around it, which decides online whether the mass that scales the residual can be trusted and acts when it cannot; dead-zone and covariance-management schemes [22], [23] stop the update on a generic excitation proxy that cannot tell hover from a maneuver (Section VII); Fisher-optimal input design [21], [25] and dual control [26] plan the excitation offline or implicitly in the cost; and recursive inertial identification [4], [5], [27] assumes excited flight. None of them evaluates online, from a closed-form signal, whether the mass is currently separable from the vertical force, and none uses that signal both to protect the estimate and to trigger the excitation. Second, the external force and the inertial parameters are coupled, so estimating them jointly can bias both. The near-hover mass–vertical-force degeneracy we use is a known structural fact in aerial estimation; existing methods, to our knowledge, do not quantify it online to decide *when* a parameter is reliable or to *act* so that it becomes identifiable. In

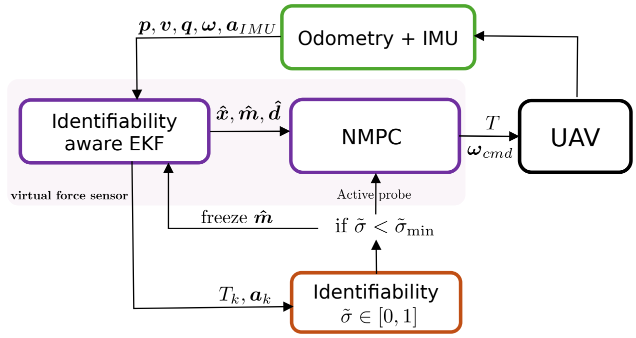

FIGURE 1. Architecture of the self-calibrating virtual force sensor: the identifiability σ gates the mass calibration and triggers the active probe; the NMPC feeds $\hat{d}$ forward.

near-hover flight the mass and the vertical external force enter the dynamics through the same channel and are not jointly identifiable, so an estimator that updates both distributes the error between them. This is the property we exploit.

This paper builds an identifiability-aware, self-calibrating virtual force sensor. The external force enters the accelerometer almost algebraically, so the sensor reads it accurately once the mass is calibrated; the mass, by contrast, is only conditionally identifiable and is confounded with a vertical force near hover. We make the calibration identifiability-aware on two levels. An extended Kalman filter (EKF) monitors an online identifiability measure and freezes the mass calibration when the parameter is unidentifiable, protecting the reading during hover where a naive filter drifts. When the mass must be recalibrated but the flight is too weakly exciting to reveal it, the model-predictive controller injects a minimal *probe*, a small bounded excursion of the position reference (not a sensor), that restores identifiability and then stops, so the sensor never relies on a parameter it cannot see. Figure 1 shows the resulting loop and the role of the identifiability signal $\tilde{\sigma}$.

The contributions of this work are the following. The bare reading of the external specific force from the accelerometer residual is a known momentum/IMU-residual technique [7], [8], [15]; our contribution is the identifiability-aware self-calibration layer that makes such a virtual force sensor reliable on a vehicle whose calibration parameter is only conditionally identifiable.

- We give the identifiability analysis of the sensor's calibration: the mass and the vertical external force are not jointly identifiable in near-hover flight, and we derive an online identifiability measure, the Schur complement of the force block in the thrust–acceleration regressor Gramian, which is the marginal Fisher information on the mass (its inverse is the Cramér–Rao bound) and quantifies when the calibration is reliable.
- We make the sensor *self-calibrating*. The EKF gates the mass calibration by this measure, with an explicit quantitative criterion, the normalized Schur complement falling below a threshold ($\tilde{\sigma} < \tilde{\sigma}_{\min}$, (18)), freezing

TABLE 1. Handling of the calibration parameter's identifiability in related approaches. "Gate signal" is what decides whether the parameter is updated; "m–d_z" asks whether the method isolates the mass–vertical-force coupling; "restores" asks whether it acts to recover identifiability.

Approach	What it measures	Gate signal	m–d_z	restores
Momentum / IMU-residual obs. [7], [8]	external force (m assumed)	none	no	no
ESO / disturbance obs. [17], [24]	lumped disturbance (m assumed)	none	no	no
Dead-zone, cov. management [22], [23]	parameter error	generic PE proxy	no	no
Fisher-optimal input design [21], [25]	information of a planned input	offline	no	offline
Dual / active-learning MPC [26]	expected future information	implicit (cost)	no	implicit
Recursive inertial ID [4], [5], [27]	inertial set, excited flight	none	partial	no
This work	σ = marginal Fisher info. on m	$\tilde{\sigma}$, online, closed form	yes	online

the mass when it is unidentifiable so the reading is protected during hover; and when the mass must be recalibrated but the flight lacks sufficient excitation, the controller injects a probe to restore identifiability. The same $\tilde{\sigma}$ *shapes* the probe, since maximizing it selects the most informative excitation within the probe family, so a single identifiability scalar governs gating, triggering, and excitation design.

The resulting sensor is characterized and validated end to end in software-in-the-loop: it matches the simulator ground truth to about $0.4\,\mathrm{N}$ RMS per axis (per-axis correlation 0.93 under continuous wind), a moving-horizon estimator does not improve on it because the reading is algebraically identifiable, and feeding the sensed force to a model-predictive controller reduces the tracking error by 34 to $80\,\%$ under wind, in real time at $100\,\mathrm{Hz}$.

This is a methodology and feasibility study in software-in-the-loop. We exercise the sensor against injected accelerometer bias, scale-factor, and thrust-mismatch errors to probe the dominant real-sensor effects, but a full hardware demonstration, on a real IMU and thrust map, or on a manipulator whose joint-torque readings give the same identifiability structure, is the natural next step and is deliberately left to a dedicated follow-up.

The rest of the paper is organized as follows. Section II presents the translational model and the measurement equations. Section III analyzes identifiability and derives the hover ambiguity and the identifiability measure. Section IV describes the identifiability-aware EKF and its mass gate. Section V presents the disturbance-rejecting controller and the active excitation. Section VI reports the experimental setup and Section VII the results. Section VIII concludes the paper.

II. SYSTEM MODEL

We model the quadrotor as a rigid body and estimate only its translational motion. The attitude is measured by the onboard estimator and enters the model as a known input, which removes the orientation states from the estimator and keeps it well-conditioned.

A. FRAMES AND NOTATION

Let $\mathcal{F}_W = \{O_W, \boldsymbol{x}_W, \boldsymbol{y}_W, \boldsymbol{z}_W\}$ be the inertial world frame and $\mathcal{F}_B = \{O_B, \boldsymbol{x}_B, \boldsymbol{y}_B, \boldsymbol{z}_B\}$ the body frame attached to the center of mass (CoM). The position of $\mathcal{F}_B$ with respect to $\mathcal{F}_W$ is $\boldsymbol{p} \in \mathbb{R}^3$ and its velocity is $\boldsymbol{v} \in \mathbb{R}^3$. The orientation of $\mathcal{F}_B$ with respect to $\mathcal{F}_W$ is the rotation matrix $\mathbf{R}(\boldsymbol{q}) \in SO(3)$, parameterized by the unit quaternion $\boldsymbol{q} = [q_0,\ \boldsymbol{q}_v^\top]^\top \in \mathbb{S}^3$, with $\mathbb{S}^3 \triangleq \{\boldsymbol{q} \in \mathbb{H} : \|\boldsymbol{q}\| = 1\}$ following the convention of [28]. The body attitude rate (body rate for short) $\boldsymbol{\omega} \in \mathbb{R}^3$ is the angular velocity of $\mathcal{F}_B$ with respect to $\mathcal{F}_W$ expressed in $\mathcal{F}_B$. We write $[\cdot]_\times$ for the skew-symmetric operator that maps a vector to the matrix form of the cross product. The thrust acts along $\boldsymbol{z}_B$, so its direction in $\mathcal{F}_W$ is $\boldsymbol{a} \triangleq \mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3$ with $\boldsymbol{e}_3 = [0,0,1]^\top$.

B. RIGID-BODY DYNAMICS

The quadrotor is a six-degree-of-freedom rigid body actuated by a collective thrust T along the body z axis and by an inner-loop body-rate controller. Its state is the position $\boldsymbol{p}$, velocity $\boldsymbol{v}$, attitude $\boldsymbol{q}$, and body rate $\boldsymbol{\omega} \in \mathbb{R}^3$. The dynamics are

$$\dot{\boldsymbol{p}} = \boldsymbol{v}, \tag{1}$$
$$\dot{\boldsymbol{v}} = -g\,\boldsymbol{e}_3 + \frac{T}{m}\mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3 + \boldsymbol{d} - c\,\|\boldsymbol{v}\|\,\boldsymbol{v}, \tag{2}$$
$$\dot{\boldsymbol{q}} = \tfrac{1}{2}\,\boldsymbol{\Omega}(\boldsymbol{\omega})\,\boldsymbol{q}, \tag{3}$$
$$\dot{\boldsymbol{\omega}} = \tfrac{1}{\tau_{rc}}\,(\boldsymbol{\omega}_{\mathrm{cmd}} - \boldsymbol{\omega}), \tag{4}$$

where g is the gravitational acceleration, $m \in \mathbb{R}_{>0}$ is the mass, and c is a quadratic drag coefficient that lumps the dominant unmodeled aerodynamics [29]. The attitude kinematics (3) use the matrix $\boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}]_\times \end{bmatrix}$, following the convention of [28], [30]. The body rate follows the first-order response (4) to the commanded rate $\boldsymbol{\omega}_{\mathrm{cmd}}$, with time constant τ_{rc} for the inner-loop rate controller, while the collective thrust T is commanded directly. The term $\boldsymbol{d} \in \mathbb{R}^3$, expressed in $\mathcal{F}_W$, is the *external specific force*, the net external force per unit mass acting at the center of mass,

$$\boldsymbol{d} \triangleq \frac{1}{m}\boldsymbol{f}_e, \tag{5}$$

where $\boldsymbol{f}_e \in \mathbb{R}^3$ is the resultant of the forces that the nominal model does not represent: aerodynamic effects beyond the drag term, wind, and any added payload. Equivalently, $\boldsymbol{d}$ is the residual between the true acceleration and the acceleration that gravity and thrust predict in (2); the experiments set the explicit drag coefficient $c = 0$, so $\boldsymbol{d}$ also lumps the aerodynamic drag. We treat $\boldsymbol{d}$ as quasi-static over the estimation window in near-hover flight, where the velocity-dependent drag is negligible and the weight of an added mass and a vertical force become indistinguishable (Section III); under fast flight $\boldsymbol{d}$ varies with velocity, which the random-walk process model tracks within its bandwidth. Thus $\boldsymbol{d}$ is a *lumped specific-force residual*: the external force (wind, payload) plus the lumped aerodynamic drag, the former dominating near hover and the latter growing with speed. The force estimate is validated against the simulator's external-force channel, and the residual drag is what the random-walk bandwidth limits at high speed (Section VII). The quantities estimated online are the mass m and the external force $\boldsymbol{d}$, which enter the translational dynamics (2); the rate-loop constant τ_{rc} is a nominal known constant of the inner loop, not estimated. To be explicit about what the sensor relies on: g is known; the drag coefficient c is not identified but set to zero, so that any drag is lumped into $\boldsymbol{d}$ and read as part of the force; τ_{rc} enters only the controller's prediction model, not the estimator; the mass is estimated; and the thrust T comes from the vehicle's thrust map, which is the one calibration the reading depends on and whose error is characterized in Section VII-H (a 10 % thrust mismatch couples almost one-to-one into the mass and barely into the force).

For estimation the attitude is treated as a known input: $\boldsymbol{q}$, and hence the thrust axis $\boldsymbol{a}$, are taken from the accurate onboard estimate, so the estimator integrates only the translational dynamics (1)–(2), with the thrust T as a known input. This avoids the quaternion states, with their re-normalization and hemisphere bookkeeping, and keeps the estimator well-conditioned over the window.

C. MEASUREMENT MODEL

The onboard sensors provide several signals at each step: the position and velocity from odometry, the attitude $\boldsymbol{q}$ and the body rate $\boldsymbol{\omega}$ from the inertial estimator and the gyroscope, and the specific force from the accelerometer. The translational estimator uses two of them and treats the attitude as an input. The first is the position–velocity measurement

$$\boldsymbol{y} = [\boldsymbol{p}^\top,\ \boldsymbol{v}^\top]^\top \in \mathbb{R}^6, \tag{6}$$

provided by the odometry, which is noisy in the simulator. The second is the accelerometer, which measures the specific force in $\mathcal{F}_B$, the non-gravitational acceleration sensed by the IMU [31]; rotated into $\mathcal{F}_W$ with the measured attitude it reads

$$\boldsymbol{s} = \mathbf{R}(\boldsymbol{q})\,\boldsymbol{a}_{\mathrm{IMU}} = \frac{T}{m}\boldsymbol{a} + \boldsymbol{d} - c\,\|\boldsymbol{v}\|\,\boldsymbol{v}, \tag{7}$$

the non-gravitational part of (2). The attitude $\boldsymbol{q}$ enters as the known input through $\boldsymbol{a} = \mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3$ rather than as a state to estimate, and the body rate $\boldsymbol{\omega}$ is not needed by the translational estimator. Because the thrust T and the attitude $\boldsymbol{q}$ are known, (7) is an algebraic measurement of $\boldsymbol{d}$: the accelerometer residual senses the external force directly, which turns the estimator into a virtual force sensor.

III. IDENTIFIABILITY AND EXCITATION

The accelerometer measurement (7) relates the unknowns to known signals, so we analyze when the mass and the external force can be recovered from it. The drag term is a known function of the measured velocity, so we move it to the left-hand side and define the compensated specific force $\tilde{\boldsymbol{s}} \triangleq \boldsymbol{s} + c\,\|\boldsymbol{v}\|\,\boldsymbol{v}$. With the inverse mass $\beta \triangleq m^{-1}$, (7) becomes

$$\tilde{\boldsymbol{s}} = T\,\boldsymbol{a}\,\beta + \boldsymbol{d} = \boldsymbol{\Phi}\,\boldsymbol{\varphi}, \qquad \boldsymbol{\Phi} = [\,T\boldsymbol{a}\ \ \mathbf{I}_3\,], \quad \boldsymbol{\varphi} = [\beta,\ \boldsymbol{d}^\top]^\top, \tag{8}$$

which is linear in the unknown $\boldsymbol{\varphi} \in \mathbb{R}^4$ with regressor $\boldsymbol{\Phi} \in \mathbb{R}^{3\times 4}$. The thrust T and the thrust axis $\boldsymbol{a}$ are known, so identifiability of $\boldsymbol{\varphi}$ is the identifiability of $(m, \boldsymbol{d})$.

The accelerometer is what makes (8) algebraic. Without it, the force $\boldsymbol{d}$ would enter only through the velocity derivative $\dot{\boldsymbol{v}}$ in (2) and would have to be recovered by differentiating the noisy odometry twice, which amplifies noise and adds a delay. Reading $\boldsymbol{d}$ from the specific force (7) removes that differentiation and yields an instantaneous algebraic measurement, which is the basis of the virtual force sensor and decouples the analysis of $\boldsymbol{d}$ from the integration of the trajectory.

The same distinction separates the reading from an energy-based one. With all parameters known, the work done by the external force is the difference between the energy delivered by the thrust and the change in the vehicle's kinetic and potential energy, and it could be read from that balance. But the balance is a scalar (it loses the direction of the force), it is an integral (the force enters as $\int \boldsymbol{d}\cdot\boldsymbol{v}\,dt$, so it lags), and it vanishes wherever the velocity does, which is exactly the hover where the force matters most. The specific-force reading (8) is a vector, instantaneous, and independent of the velocity.

The estimator augments the position and velocity with the mass and the force. The position and velocity are measured by the odometry (6) at every step, so the recoverability of the parameters reduces to the *identifiability* of $(m, \boldsymbol{d})$ from the accelerometer model (8) accumulated over the window, a persistency-of-excitation condition on the regressor rather than a full nonlinear observability test of the augmented state. We therefore speak of *identifiability* throughout: the property in question is that of the parameters $(m, \boldsymbol{d})$ in the regressor, not the observability of the state, although the two coincide here because the position and the velocity are measured directly; a nonlinear observability treatment of the inertial parameters of a rigid body can be found in [27]. This identifiability determines when the mass and the force can be separated and is what we analyze and exploit.

A single sample gives three equations for four unknowns, so $\boldsymbol{\varphi}$ is recovered only by accumulating samples. Over a window of N samples the information is the Gramian $\mathbf{G} = \sum_{k=1}^{N} \boldsymbol{\Phi}_k^\top \boldsymbol{\Phi}_k$, which has the block form

$$\mathbf{G} = \begin{bmatrix} \sum_k T_k^2 & \sum_k T_k \boldsymbol{a}_k^\top \\ \sum_k T_k \boldsymbol{a}_k & N\,\mathbf{I}_3 \end{bmatrix}, \tag{9}$$

where $\|\boldsymbol{a}_k\| = 1$ was used. The force block $N\mathbf{I}_3$ is always nonsingular, so given the mass, $\boldsymbol{d}$ is recovered whenever the window is nonempty; the joint degeneracy that remains is confined to the mass–vertical-force pair at hover (Proposition 1). The mass enters only through β, which is identifiable if and only if the Schur complement of the force block is positive,

$$\sigma \triangleq \sum_k T_k^2 - \tfrac{1}{N} \left\| \sum_k T_k \boldsymbol{a}_k \right\|^2 > 0. \tag{10}$$

The scalar σ is the dispersion of the thrust *vectors* $T_k\boldsymbol{a}_k$ about their mean, and it becomes positive through a change in the thrust magnitude T_k, a change in its direction $\boldsymbol{a}_k$, or both; it is zero only when the vectors are identical across the window, which is the case at hover. Its dimension is thrust squared (newtons squared; the italic N denotes the window length), so its absolute scale grows with the thrust level of the maneuver. To obtain a dimensionless, maneuver-comparable measure we normalize it by the thrust energy in the window,

$$\tilde{\sigma} \triangleq \frac{\sigma}{\sum_k T_k^2} \in [0, 1], \tag{11}$$

the fraction of the thrust energy that lies in the dispersion: $\tilde{\sigma} = 0$ at hover and $\tilde{\sigma} \to 1$ when the thrust vectors are maximally spread. Normalizing by the absolute thrust level removes the dominant source of the threshold's maneuver dependence, so a single $\tilde{\sigma}_{\min}$ classifies the regime across maneuvers far better than the raw σ (Section VII-I). The normalization also discards information that the bound below retains: two windows with the same $\tilde{\sigma}$ but different thrust levels, or different accelerometer noise, carry different absolute Fisher information. We therefore keep the two roles apart throughout: $\tilde{\sigma}$ is the dimensionless *classification* signal that decides the regime, and σ/r_s is the *information content* that lower-bounds the variance; Section VII-I reports both side by side for every regime flown.

Proposition 1. *In stationary level hover, where $\boldsymbol{a}_k = \boldsymbol{e}_3$ and $T_k = \bar{T}$ are constant over the window, the inverse mass β is not identifiable ($\sigma = 0$). Equivalently, only the combination $\bar{T}\beta + d_z$ is identifiable, while β and the vertical force d_z are not separately identifiable.*

Proof. With $\boldsymbol{a}_k = \boldsymbol{e}_3$ and $T_k = \bar{T}$ we have $\sum_k T_k^2 = N\bar{T}^2$ and $\sum_k T_k\boldsymbol{a}_k = N\bar{T}\boldsymbol{e}_3$, so $\sigma = N\bar{T}^2 - \frac{1}{N}N^2\bar{T}^2 = 0$, and β is not identifiable by (10). The structure is explicit in (8): the vertical row reads $\tilde{s}_z = \bar{T}\beta + d_z$, a single equation in the two unknowns β and d_z, whereas the horizontal rows read $\tilde{s}_{x,y} = d_{x,y}$ since $\boldsymbol{a} = \boldsymbol{e}_3$. Hence d_x and d_y are identifiable while β and d_z collapse onto their sum. □

Remark 1. *The ambiguity is confined to the vertical axis and to the pair (m, d_z): a mass error and a vertical force produce the same accelerometer residual at hover. The joint ambiguity is confined to the mean thrust direction, which at and near hover is vertical, so the horizontal components d_x and d_y remain identifiable there, which is why the external force is recovered in the directions where wind dominates.*

Away from hover the thrust axis tilts and its magnitude varies, so the vectors $T_k\boldsymbol{a}_k$ disperse and $\sigma > 0$; the window is then persistently exciting for the mass. Since the Cramér–Rao lower bound on the mass estimate scales with σ^{-1} (derived in (12) below), a more aggressive trajectory admits a better-conditioned estimate, while the force, conditioned by the block $N\mathbf{I}_3$ given the mass, is read with the same accuracy in every regime with an accurate held mass. The mass is thus identifiable under motion and degenerate only at hover, which Section VII confirms by identifying it accurately across the range of flight speeds.

The scalar σ is therefore an online measure of how well the mass can be identified, read directly from the known thrust and attitude over the window with no extra sensing. It also has an estimation-theoretic meaning, which we state as a proposition because the gate and the probe of Section V rest on it.

Proposition 2 (Cramér–Rao bound on the mass)**.** *Let the accelerometer noise be zero-mean with isotropic covariance $r_s\mathbf{I}_3$ and let $\boldsymbol{d}$ be constant over the window. Then the marginal Fisher information on the inverse mass β, after profiling out the jointly estimated force $\boldsymbol{d}$, is σ/r_s, and any unbiased estimate satisfies*

$$\operatorname{var}(\hat{\beta}) \;\geq\; r_s\,[\mathbf{G}^{-1}]_{11} \;=\; r_s\,\sigma^{-1}. \tag{12}$$

Proof. The joint Fisher information of $(\beta, \boldsymbol{d})$ over the window is $\mathbf{G}/r_s$, with $\mathbf{G}$ the Gramian (9). By the block-inverse formula, the $(1,1)$ entry of $\mathbf{G}^{-1}$ is the inverse of the Schur complement of the force block, $[\mathbf{G}^{-1}]_{11} = \sigma^{-1}$, and the Cramér–Rao inequality gives (12). □

The bound diverges as $\sigma \to 0$ at hover and tightens as the maneuver disperses the thrust vectors. It is the batch least-squares bound under the stated assumptions, the limit that the recursive filter approaches under excitation, and we use it as a relative identifiability measure rather than an absolute uncertainty. It is stated on the inverse mass β; since $\operatorname{var}(\hat{m}) \approx m^4 \operatorname{var}(\hat{\beta})$ for small errors, the bound on the mass itself scales with the same σ^{-1}.

Proposition 2 holds $\boldsymbol{d}$ constant over the window, whereas the filter of Section IV models it as a random walk of per-step variance q_d, and the applied disturbance contains wind steps and, at speed, a velocity-dependent drag. Under the random-walk model the nuisance is a trajectory rather than a single vector, and the exact marginal Fisher information on β is a weighted Schur complement $\sigma_{\text{eff}} \leq \sigma$, derived at the end of this section, which reduces to σ as $q_d \to 0$. Section VII-I evaluates it on logged flight data: for the implemented $q_d = 10^{-2}$, a drift of about $1\,\mathrm{m\,s^{-2}}$ per second (half the applied wind), the information in a $1\,\mathrm{s}$ window is $12\,\%$ of the constant-force value, and the ratio decreases with the window length because a longer window gives the random walk more freedom to absorb the slow variations of $T\boldsymbol{a}$. Proposition 2 is thus an optimistic bound for the implemented filter; Section VII-I shows that the random-walk bound accounts for most of the gap between it and the empirical spread of the estimate. For the disturbance profiles (steps and sinusoid) and the range of q_d evaluated, the gate decision is unaffected by the approximation: it thresholds the dimensionless $\tilde{\sigma}$ as a regime classifier with an empirical margin, and at hover σ_{eff} stays at the noise floor for every q_d tested, so the separation between the regimes survives (Section VII-I); we did not sweep the physical bandwidth of the disturbance separately.

Exact information under the random-walk force model. Stack the compensated specific forces of the window as $\boldsymbol{S} = [\tilde{\boldsymbol{s}}_1^\top, \ldots, \tilde{\boldsymbol{s}}_N^\top]^\top \in \mathbb{R}^{3N}$, and let $\boldsymbol{t} = [T_1\boldsymbol{a}_1^\top, \ldots, T_N\boldsymbol{a}_N^\top]^\top$ and $\mathbf{M} = [\mathbf{I}_3, \ldots, \mathbf{I}_3]^\top \in \mathbb{R}^{3N\times 3}$. With the force evolving as $\boldsymbol{d}_k = \boldsymbol{d}_{k-1} + \boldsymbol{w}_k$, $\boldsymbol{w}_k \sim \mathcal{N}(\mathbf{0}, q_d\mathbf{I}_3)$, and accelerometer noise $\boldsymbol{n}_k \sim \mathcal{N}(\mathbf{0}, r_s\mathbf{I}_3)$, the window model (8) reads

$$\begin{aligned} \boldsymbol{S} &= \boldsymbol{t}\,\beta + \mathbf{M}\,\boldsymbol{d}_1 + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \\ \boldsymbol{\Sigma} &= r_s\mathbf{I}_{3N} + q_d\,\mathbf{C} \otimes \mathbf{I}_3, \end{aligned} \tag{13}$$

with $[\mathbf{C}]_{kl} = \min(k, l) - 1$, where the increments have been marginalized into the covariance and the initial force $\boldsymbol{d}_1$ is a nuisance parameter with a flat prior. The Fisher information of $(\beta, \boldsymbol{d}_1)$ is

$$\mathbf{J} = \begin{bmatrix} \boldsymbol{t}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{t} & \boldsymbol{t}^\top\boldsymbol{\Sigma}^{-1}\mathbf{M} \\ \mathbf{M}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{t} & \mathbf{M}^\top\boldsymbol{\Sigma}^{-1}\mathbf{M} \end{bmatrix}, \tag{14}$$

and the marginal information on β is its Schur complement,

$$\begin{aligned} \frac{\sigma_{\text{eff}}}{r_s} &\triangleq \boldsymbol{t}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{t} \\ &\quad - \boldsymbol{t}^\top\boldsymbol{\Sigma}^{-1}\mathbf{M}\left(\mathbf{M}^\top\boldsymbol{\Sigma}^{-1}\mathbf{M}\right)^{-1}\mathbf{M}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{t}. \end{aligned} \tag{15}$$

For $q_d = 0$, $\boldsymbol{\Sigma} = r_s\mathbf{I}_{3N}$ and (15) reduces to σ/r_s of Proposition 2. For $q_d > 0$, $\boldsymbol{\Sigma}^{-1}$ down-weights the slow variations of $\boldsymbol{t}$, those a random walk can follow, so $\sigma_{\text{eff}} \leq \sigma$; the fast excitation of the probe is the part that survives. Evaluating (15) is a $3N \times 3N$ solve, about $0.7\,\text{ms}$ at $N = 100$ when evaluated numerically, so it could replace σ in the gate; since the gate thresholds the dimensionless ratio with margin, we retain the closed-form $\tilde{\sigma}$.

For *gating* we use the dimensionless normalization $\tilde{\sigma}$ (11), which normalizes by the thrust level so that a single threshold $\tilde{\sigma}_{\min}$ holds across maneuvers; σ carries the information content and $\tilde{\sigma}$ carries the decision. The threshold is set with margin between the hover and maneuver values of $\tilde{\sigma}$ (Section VII-I), and Proposition 2 translates it into the best resolution that the admitted data could give. Over a window of N samples near hover thrust, $\sum_k T_k^2 \approx N(mg)^2$, so for a window that just opens the gate, $\sigma = \tilde{\sigma}_{\min}N(mg)^2$, the Cramér–Rao bound reads

$$\frac{\operatorname{std}(\hat{m})}{m} \gtrsim \frac{1}{g}\sqrt{\frac{r_s}{\tilde{\sigma}_{\min}N}}. \tag{16}$$

With the parameters used in the experiments ($\tilde{\sigma}_{\min} = 0.009$, $N = 100$, $r_s = 0.5$; Section VI) this is $7.6\,\%$ for one second of data and $2.4\,\%$ after ten. The bound is a floor, not a guarantee: for a one-second window at or below the threshold, even an efficient unbiased estimator could not attain a standard deviation smaller than $7.6\,\%$, which is of the scale of the $5\,\%$ to $7\,\%$ hover bias the gate guards against (Section VII). Under the same ideal constant-force assumptions, the floor crosses the $3.3\,\%$ residual observed in the windy hover after approximately $5.3\,\text{s}$ of accumulated admissible data and reaches $2.4\,\%$ after ten seconds. These figures characterize only the best-case variance permitted by the information; they neither predict nor upper-bound the realized estimation error. The estimator uses $\tilde{\sigma}$ to decide when to trust the mass, and the controller of Section V uses it to decide when to excite.

TABLE 2. Identifiability regimes (Proposition 1). The "degraded" d_z entry is the joint case; with an accurate held mass it is recovered (Section VII).

Quantity	Hover	Maneuver
mass m	unidentifiable ($\sigma \approx 0$)	identifiable (σ large)
d_x, d_y	identifiable	identifiable
d_z	degraded (m–d_z)	identifiable

TABLE 3. Estimator and controller parameters.

Block	Parameter	Value
Plant / sim	mass, rate τ_{rc}^{-1}	$1.05\,\text{kg}$, $18\,\text{s}^{-1}$
Drag	c (estimator)	0 (lumped into $\hat{\boldsymbol{d}}$)
Odometry noise	injected on p, v	MuJoCo, released config
Wind	profile, amplitude	step / sine, $\sim 2\,\text{N}$
EKF meas. var.	$r_p,\ r_v,\ r_s$	$0.02,\ 0.1,\ 0.5$
EKF process	$q_p,\ q_v,\ q_m,\ q_d$	$10^{-6}, 10^{-3}, 10^{-5}, 10^{-2}$
Mass gate	window, dwell t_d, $\tilde{\sigma}_{\min}$	$1.0\,\text{s}$, $0.25\,\text{s}$, 0.009
Smoothing	$\alpha_m,\ \alpha_d$	$0.995,\ 0.92$
Active probe	budget ρ, f_ℓ, f_z, optimal	$0.80\,\text{m}$, $0.5\,\text{Hz}$, $1.0\,\text{Hz}$, vertical
	withdrawal t_p (gate-open time)	$15\,\text{s}$
NMPC	N_c, horizon	31, $1.5\,\text{s}$
NMPC weights	$\mathbf{Q}_p, \mathbf{Q}_a, \mathbf{Q}_v$	$180,\ 20,\ 30$
	$\mathbf{R}_u$	$\operatorname{diag}(0.3, 5, 5, 5)$
NMPC limits	$T_{\max},\ \omega_{\max}$	$49\,\text{N}$, $20\,\text{rad}\,\text{s}^{-1}$

Proposition 1 has a direct operational reading, summarized in Table 2. The external force is recoverable in every regime given an accurate held mass, and is recovered by any recursive estimator that uses the accelerometer residual; the mass is recoverable only when the flight excites it (σ large), and is then identified jointly with the force. This motivates the two elements of our method: an estimator that updates the mass only when σ is large (Section IV), and a controller that creates the excitation when the mass must be identified but the flight lacks the excitation to reveal it (Section V).

IV. IDENTIFIABILITY-AWARE ESTIMATION

We estimate the translational state and the parameters of Section II with an extended Kalman filter (EKF) whose state augments the position and velocity with the mass and the external force,

$$\boldsymbol{x} = [\boldsymbol{p}^\top,\ \boldsymbol{v}^\top,\ m,\ \boldsymbol{d}^\top]^\top \in \mathbb{R}^{10}. \tag{17}$$

The known inputs at each step are the thrust T and the thrust axis $\boldsymbol{a}$. The position and velocity follow (1)–(2), while the mass and the external force have no dynamics of their own and evolve as random walks with diagonal process covariance $\mathbf{Q} = \operatorname{diag}(q_p\mathbf{I}_3, q_v\mathbf{I}_3, q_m, q_d\mathbf{I}_3)$ (Table 3). Treating the attitude as a known input keeps the filter free of quaternion states and their log-map singularity; the body rate is not needed by the translational estimator.

A. PREDICTION AND UPDATE

Each step propagates the dynamics, $\hat{\boldsymbol{x}}^- = \boldsymbol{f}(\hat{\boldsymbol{x}})$ and $\mathbf{P}^- = \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q}$ with $\mathbf{F} = \partial\boldsymbol{f}/\partial\boldsymbol{x}$, and applies two measurement updates. The odometry (6) is a linear update through $\mathbf{H} = [\mathbf{I}_6\ \mathbf{0}]$ with covariance $\mathbf{R}$. The accelerometer (7) is a nonlinear update with model $\boldsymbol{g}(\boldsymbol{x}) = \frac{T}{m}\boldsymbol{a} + \boldsymbol{d} - c\ \|\boldsymbol{v}\|\ \boldsymbol{v}$, Jacobian $\mathbf{H}_s = \partial\boldsymbol{g}/\partial\mathbf{x}$, and covariance $\mathbf{R}_s$; this residual is the term that drives $\boldsymbol{d}$ and makes the filter a virtual force sensor. Both updates use the standard EKF gain $\mathbf{K} = \mathbf{P}^-\mathbf{H}^\top(\mathbf{H}\mathbf{P}^-\mathbf{H}^\top + \mathbf{R})^{-1}$, and the mass and force are clipped to physical bounds after each step.

B. VALIDITY OF THE LINEARIZATION

The EKF linearizes the model around the current estimate, so it is worth stating how nonlinear the model actually is. The dynamics (1)–(2) and the measurements (6), (7) are linear in the position, the velocity, and the force; the attitude, which carries the rotational nonlinearity, enters as a known input rather than as a state; and the drag is not estimated ($c = 0$). The only nonlinear dependence on the state is the factor T/m, through which the Jacobian entry $-T\boldsymbol{a}/\hat{m}^2$ is the first-order expansion of $1/m$ about $\hat{m}$, with a relative curvature error of $(\delta m/m)^2$: below $1\,\%$ for a $10\,\%$ mass error, and $18\,\%$ for the deliberately wrong $0.60\,\mathrm{kg}$ prior of the identification experiment of Section VII, which the filter still recovers because the error shrinks with every update. The linearization is therefore not the limiting factor of this estimator, and the matched comparison with a moving-horizon estimator, which optimizes the full nonlinear window without local linearization, confirms it (Section IV-D). The situations where the formulation is less adequate are of a different kind. Under rapid rotation the reading depends on rotating the accelerometer with the attitude sampled at the same instant: a latency δt between the IMU and the attitude estimate produces an error of order $g\,\omega\,\delta t$ ($0.5\,\mathrm{m\,s^{-2}}$ at $\omega = 5\,\mathrm{rad\,s^{-1}}$ and $\delta t = 10\,\mathrm{ms}$), which is a synchronization requirement on the sensors, not a linearization issue. And because $\boldsymbol{d}$ is expressed in the world frame, a disturbance that is fixed to the body, such as the aerodynamic drag lumped into $\hat{\boldsymbol{d}}$ at speed, must be tracked by the random walk as the attitude turns it, which is the bandwidth limitation already noted in Section II; a body-frame force state, or an unscented or iterated update, would be the drop-in changes if either effect dominated on a given platform.

C. THE MASS GATE

A plain EKF updates every state at every step, including the mass when it is unidentifiable. Under the hover ambiguity of Proposition 1 this attributes a vertical force to a mass change and corrupts the estimate. The identifiability-aware filter instead conditions the mass update on the identifiability $\tilde{\sigma}$ of (11), computed online over a sliding window from the known thrust and attitude; the mass is declared unidentifiable, quantitatively, when $\tilde{\sigma}$ falls below the threshold $\tilde{\sigma}_{\min}$ (Table 3). When the mass is not persistently excited, the corresponding row of the Kalman gain is set to zero and its process noise is suppressed,

$$\mathbf{K}_m \leftarrow \mathbf{0}, \quad q_m \leftarrow 0 \qquad \text{while} \quad \tilde{\sigma} < \tilde{\sigma}_{\min}. \tag{18}$$

With $q_m = 0$ and the mass having no dynamics of its own ($F_{mm} = 1$), the prediction leaves the mass variance unchanged, $P_m^- = P_m$, so while the gate is closed the mass neither drifts nor inflates: it is held exactly at its last well-identified value with its uncertainty. The realized hold is therefore as good as the gate's closure: in still hover the gate stays shut and the mass is held exactly, whereas a windy hover intermittently re-excites $\tilde{\sigma}$ and re-opens the gate, leaving the residual movement reported in Section VII. Suppressing the process noise is what makes this hold; were the variance left to grow, a transient spike in $\tilde{\sigma}$ would re-open the update with a large gain and let a single sample move the mass. To reject such spikes the gate re-opens the update only after $\tilde{\sigma}$ has stayed above $\tilde{\sigma}_{\min}$ for a dwell time t_d (Table 3), not on an isolated sample; this separates the sustained excitation of a maneuver or probe from a momentary gust. The force update is never gated, since, given the held mass, $\boldsymbol{d}$ is identifiable throughout (Section III).

D. CHOICE OF ESTIMATOR: EKF VERSUS MOVING HORIZON

The accelerometer makes $\boldsymbol{d}$ an algebraic measurement (8), so the force needs no temporal smoothing to be recovered, and the mass, once excited, is identified by the recursive update alone. A moving-horizon estimator over the same state and measurements was implemented for comparison and did not improve the estimates (Section VII): the window adds cost without a benefit when the informative measurement is instantaneous and the recursive filter already carries the full measurement history. We therefore retain the EKF and place the contribution in the identifiability logic rather than in the estimator class.

V. CLOSED-LOOP USE: CONTROL AND ACTIVE CALIBRATION

The virtual sensor is used in closed loop by a nonlinear model-predictive controller (NMPC) that serves two roles: it rejects the sensed disturbance by feedforward (the application), and it injects the active probe that recalibrates the mass when needed (the self-calibration of Section V-B). We first describe the controller, then the probe.

The estimated force closes the loop through this NMPC, which tracks a reference trajectory while rejecting the disturbance by feedforward. The controller keeps the full pose, so its state is the position, velocity, attitude, and body rate, $\boldsymbol{x}_c = [\boldsymbol{p}^\top, \boldsymbol{v}^\top, \boldsymbol{q}^\top, \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{13}$, and its input is the collective thrust and the commanded body rate, $\boldsymbol{u} = [T,\ \boldsymbol{\omega}_{\mathrm{cmd}}^\top]^\top \in \mathbb{R}^4$. The prediction model is the rigid-body dynamics (1)–(4) with the estimated mass and the estimated disturbance injected,

$$\dot{\boldsymbol{v}} = -g\,\boldsymbol{e}_3 + \frac{T}{\hat{m}}\mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3 + \hat{\boldsymbol{d}}, \tag{19}$$

so that $\hat{\boldsymbol{d}}$ enters as an additive specific-force feedforward that cancels the external force; the rate loop keeps the form (4) with the nominal time constant τ_{rc}. The prediction model carries no explicit drag term; because the estimator runs with

$c = 0$, the aerodynamic drag is lumped into $\hat{\boldsymbol{d}}$ and is therefore compensated through the same feedforward. The disturbance estimate is smoothed by an exponential moving average before injection (the mass estimate is smoothed likewise, with the constants of Table 3). The smoothing is not a deficiency of the estimator but a separation of roles: the filter's q_d is set for the bandwidth of the force reading, so its output retains sample-to-sample noise at 100 Hz that the controller would otherwise pass straight into the thrust command; the moving average ($\alpha_d = 0.92$, a time constant of 0.12 s) is a low-pass on the actuation path that can be tuned independently of the estimation bandwidth. The mass is smoothed more heavily ($\alpha_m = 0.995$, 2 s) because it sets the hover equilibrium of the prediction model, and a step in it would step the commanded thrust. In the closed-loop experiments the online update is the force alone: the mass in the prediction model is held at its identified value (here the known 1.05 kg), so it is the feedforward $\hat{\boldsymbol{d}}$ that closes the loop, while the mass calibration protects the force reading rather than retuning the controller. Because the gate acts only on the mass row of the filter, the force estimate of the gated and the plain filter coincide, and the injected $\hat{\boldsymbol{d}}$ is that common reading.

A. COST AND CONSTRAINTS

Over a horizon of N_c nodes the controller minimizes

$$J = \sum_{k=0}^{N_c-1} \Big(\|\boldsymbol{p}_k - \boldsymbol{p}_k^{\mathrm{ref}}\|^2_{\mathbf{Q}_p} + \|\boldsymbol{v}_k - \boldsymbol{v}_k^{\mathrm{ref}}\|^2_{\mathbf{Q}_v} + \|\boldsymbol{e}_q(\boldsymbol{q}_k)\|^2_{\mathbf{Q}_a} + \|\boldsymbol{u}_k - \boldsymbol{u}_k^0\|^2_{\mathbf{R}_u} \Big) + \|\boldsymbol{e}_N\|^2_{\mathbf{Q}_N}, \tag{20}$$

where $\boldsymbol{p}^{\mathrm{ref}}$ and $\boldsymbol{v}^{\mathrm{ref}}$ are the reference position and velocity, $\boldsymbol{e}_q = \mathrm{Log}(\boldsymbol{q}^{-1} \otimes \boldsymbol{q}^{\mathrm{ref}}) \in \mathbb{R}^3$ is the attitude error through the quaternion logarithm, with the standard hemisphere correction, and the input is regularized around the hover equilibrium $\boldsymbol{u}^0 = [\hat{m}g,\ \mathbf{0}^\top]^\top$ so that a wrong nominal mass does not bias the regularization. The velocity reference $\boldsymbol{v}^{\mathrm{ref}}$ is a feedforward term that removes the tracking lag at high speed; $\mathbf{R}_u$ weights the input deviation. The terminal error $\boldsymbol{e}_N$ stacks the position, velocity, and attitude errors at stage N_c, weighted by $\mathbf{Q}_N = \mathrm{diag}(\mathbf{Q}_p, \mathbf{Q}_v, \mathbf{Q}_a)$. The terms of (20) carry different units, and the weights are their normalization: with the values of Table 3, a position error of 0.1 m, a velocity error of 0.25 m s^{-1}, an attitude error of 0.3 rad, and a thrust deviation of 2.5 N each contribute about 1.8 to the stage cost, so no single term dominates at the errors the controller operates at. The inputs are bounded by $0 \le T \le T_{\max}$ and $\|\boldsymbol{\omega}_{\mathrm{cmd}}\|_\infty \le \omega_{\max}$, with $T_{\max} = 49$ N ($\approx 4.8mg$) and $\omega_{\max} = 20$ rad s^{-1}. The control law is the receding-horizon rule: at each step the problem is solved from the current estimate and the first input of the optimal sequence is applied,

$$\boldsymbol{u}_k = \boldsymbol{u}_0^\star(\hat{\boldsymbol{x}}_k, \hat{m}, \hat{\boldsymbol{d}}_k), \tag{21}$$

after which the horizon shifts by one step and the problem is solved again.

The problem is solved with sequential quadratic programming under the real-time iteration scheme, HPIPM, and a Gauss–Newton Hessian within acados [32], sharing the framework with the estimator. It uses $N_c = 31$ nodes over a 1.5 s horizon at 100 Hz, with a per-step solve time below 1 ms. All timings in this paper are measured on a laptop-class processor (Intel Core Ultra 9 275HX, one core per solver), not on an embedded flight computer; the method has not been validated on embedded hardware.

B. ACTIVE EXCITATION

The gate of Section IV-C protects a mass that has already been identified, but it cannot create the information the flight does not provide: in sustained hover the mass stays unidentifiable and the estimate is frozen at the prior it started from. When an up-to-date mass is required and the identifiability is low, the controller closes this gap by actively exciting the unidentifiable direction. The same identifiability that gates the estimate now *shapes* the excitation: by (12) the Schur complement is the Fisher information on the mass, so the probe that maximizes it (approximately, its dimensionless form $\tilde{\sigma}$, since the probe perturbs the thrust energy only to second order about $T \approx mg$) per unit deviation maximizes the corresponding best-case information on the mass. We do not solve the continuous optimal-input-design problem [21], nor the flight-test input design of [25]; instead we maximize $\tilde{\sigma}$ over a small two-parameter probe family by an empirical sweep, the most informative probe within that family. Proposition 1 identifies the two levers: a vertical bob of amplitude δ_z that modulates the thrust magnitude T, and a lateral motion of amplitude δ_ℓ that tilts the thrust axis $\boldsymbol{a}$. Both disperse the regressor $T\boldsymbol{a}$, so the probe is

$$\boldsymbol{p}^{\mathrm{ref}} \leftarrow \boldsymbol{p}^{\mathrm{ref}} + \big[\delta_\ell(\cos\omega_\ell t - 1),\ \delta_\ell \sin\omega_\ell t,\ \delta_z \sin\omega_z t\big]^\top \quad \text{while } \tilde{\sigma} < \tilde{\sigma}_{\min}, \tag{22}$$

with $\omega_\ell = 2\pi f_\ell$ and $\omega_z = 2\pi f_z$ (Table 3), and we choose the (δ_z, δ_ℓ) mix that maximizes $\tilde{\sigma}$ subject to a fixed amplitude budget $\delta_z^2 + \delta_\ell^2 \le \rho^2$, identified empirically in Section VII. The probe is triggered only while the mass is both needed and unidentifiable, and it is withdrawn once the gate has been open for a total of t_p seconds (Table 3). The value $t_p = 15$ s is an empirical choice, validated in the five repetitions of Section VII, where the estimate has settled by then; for reference, the constant-force bound (16) for the information the probe accumulates in that time is about 1 %, the order of the residual observed. The gate then again protects the mass. This is the active counterpart of the gate.

C. REMARKS ON CLOSED-LOOP BEHAVIOR

In the evaluated runs, the model error reaching the controller remains small. The uncompensated force is the estimation residual $\boldsymbol{d} - \hat{\boldsymbol{d}}$, and the mass enters only as the multiplicative thrust scaling $\frac{\hat{m}}{m}$. Equation (12) lower-bounds the variance attainable where the mass is identifiable; where it is not, the gate freezes the mass at its last identified value. Neither statement is an upper bound on the realized estimation error. A formal stability or boundedness analysis of the coupled estimator–controller loop is left to future work.

VI. EXPERIMENTAL SETUP

We evaluate the virtual sensor and its self-calibration in software-in-the-loop; this section describes the simulation environment, the flights and disturbance, the baselines, and the metrics.

A. SIMULATION ENVIRONMENT

All experiments run in software-in-the-loop with the MuJoCo physics engine [33]. The quadrotor has a mass of $1.05\,\mathrm{kg}$, the sum of the body geometries in the model description, which we take as the ground truth for the mass estimate. The odometry is published with injected noise on position and velocity, so the estimator is exercised on noisy measurements, and the simulator exposes the true external force applied to the airframe on a dedicated channel, which serves as ground truth for the force estimate. The estimator, controller, and the scripts that reproduce every figure are provided in the repository given at the end of the paper, together with a data dictionary that maps each logged column to its physical meaning and the source file for every figure and table. The parameters are listed in Table 3.

B. CONTROLLER, TRAJECTORIES, AND DISTURBANCE

The vehicle is flown by the NMPC of Section V at $100\,\mathrm{Hz}$, the same rate as the estimator. Two references are used: a stationary hover and a Lissajous figure-eight flown at peak speeds from 2.1 to $10.2\,\mathrm{m\,s^{-1}}$, which span the excitation levels of the identifiability analysis. The external force is a *deterministic* per-axis step profile of about $2\,\mathrm{N}$, commanded by the controller on the simulator's external-force channel; the simulator applies it and re-publishes the *applied* wrench as ground truth. Because the profile is defined in code, the disturbance is identical across runs and its onset is controlled (it is armed only once the vehicle has settled, after the verified reset), which makes every experiment reproducible. The force estimate, by contrast, is read from the accelerometer residual, independent of the commanded value, so the comparison against ground truth is not circular. Sustained steps keep the disturbance within the bandwidth of the momentum-based reading, examined in Section VII. For the rejection study the same disturbance is applied to the runs with and without the feedforward.

C. BASELINE

The baseline is the *plain EKF*, the same filter of Section IV but without the mass gate, run on the identical measurement stream. This ablation isolates the effect of the identifiability logic, since the two filters differ only in the gate. As a check on the estimator class, we also ran a moving-horizon estimator over the same state and measurements; it did not improve the estimates (Section IV-D) and is not carried as a separate baseline.

TABLE 4. EKF estimation against trajectory speed (N=10; 9 in hover): mass estimate and per-axis force correlation with the ground truth. The force is read accurately throughout; the (ungated) mass is biased and noisy in hover and accurate under excitation.

Peak speed $[\mathrm{m\,s^{-1}}]$	$\hat{m}$ [kg]	corr d_x/d_y	corr d_z
hover	1.101 ± 0.035	0.98/0.98	0.97
2.1	1.083 ± 0.027	0.97/0.98	0.98
3.8	1.049 ± 0.028	0.97/0.98	0.95
6.1	1.051 ± 0.010	0.97/0.97	0.91
8.2	1.056 ± 0.003	0.96/0.95	0.94
10.2	1.053 ± 0.002	0.95/0.94	0.93

D. METRICS

The virtual force sensor is assessed by the Pearson correlation between the estimated force and the ground truth, per axis. Mass identification is assessed by the error relative to the ground truth and its spread across repetitions. Disturbance rejection is assessed by the position tracking error, reported as the root-mean-square error with and without the feedforward under the same wind, and its significance is tested with a one-sided Mann–Whitney U test at each speed. Each condition uses five repetitions with and five without the feedforward, so the pooled estimation statistics are $N = 10$ per speed; the hover feedforward case has four (one run failed the reset check and was dropped), making it $N = 4$ for the hover rejection point and $N = 9$ for the pooled hover estimation. We report mean $\pm$ std and the per-step solve time. The gate and active-excitation results are each reported over five repetitions with independent noise realizations (and wind realizations for the gate runs; the active-excitation runs are wind-free), shown as mean $\pm$ std; the disturbance-free mass-identification result is a single representative run.

VII. RESULTS

We present the results in the order of the contributions: the external force, the mass identification and its hover limit, the gate, the active excitation, the disturbance rejection, robustness, and a summary comparison against both baselines.

A. THE VIRTUAL FORCE SENSOR

The accelerometer makes the force algebraic (8), so the EKF recovers it without any temporal smoothing. We report four complementary measures so that no single one is misleading: the absolute error, the time-domain force estimate (Fig. 2), the per-axis Pearson correlation against the step ground truth (Fig. 3), and the correlation under a continuously varying disturbance (Section VII-G). The absolute error is $0.4\,\mathrm{N}$ root-mean-square per axis on a $\sim\!2\,\mathrm{N}$ disturbance, about $20\,\%$ of the applied force; the calibration curve below shows this residual is dominated by the bandwidth of the random-walk force model, not by a scale error. The step correlation is 0.91 to 0.98 across speeds (inflated by the step transitions, hence reported alongside the absolute error); and under a sinusoidal wind the correlation is 0.93 (Section VII-G), the most demanding case. The high-speed rows in Table 4 are also conservative in a specific sense: the ground-truth channel

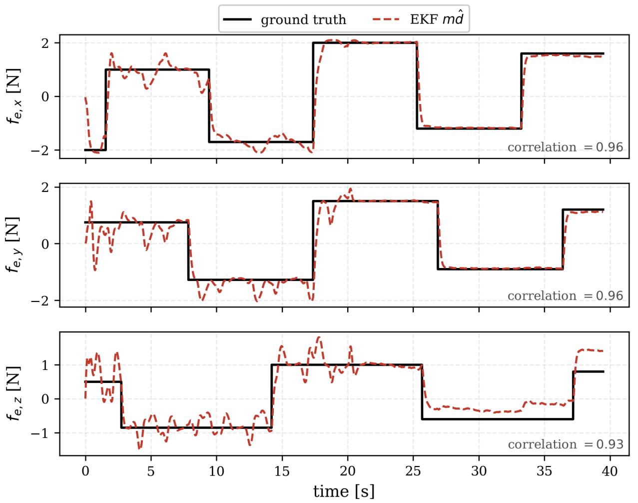

FIGURE 2. Virtual force sensor in the time domain (representative run): $m\hat{d}$ tracks the applied wind steps on all three axes.

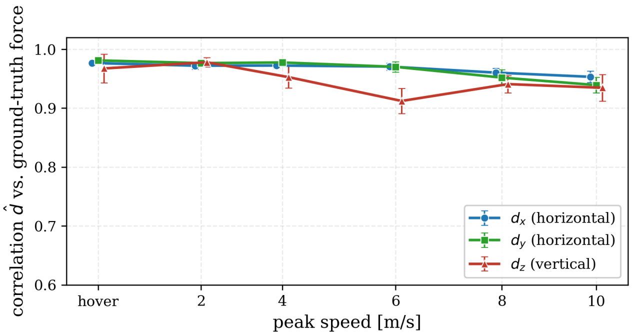

FIGURE 3. Per-axis force correlation against excitation (N=10; 9 in hover; step ground truth).

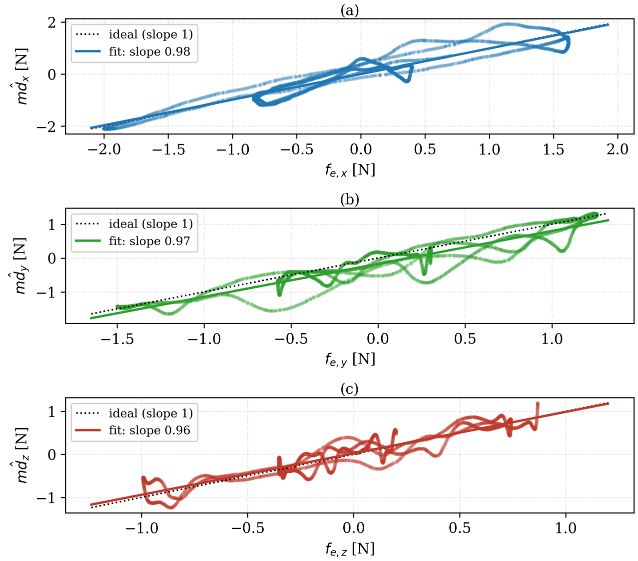

FIGURE 4. Calibration curve under a sinusoidal wind: estimated force against applied force for the (a) x, (b) y, and (c) z axes, with the linear fit (slopes 0.98, 0.97, 0.96; R^2 0.90, 0.89, 0.87) and the ideal unit slope.

contains only the applied wind, while the aerodynamic drag is additionally lumped into $\hat{\boldsymbol{d}}$ (c=0 in the estimator), so the correlations at speed mix the two; near hover, where the drag vanishes, the reading is validated cleanly. This is also why the vertical correlation in Table 4 falls from 0.97 to 0.98 near hover to 0.91 to 0.95 at speed while the horizontal ones barely move: the figure-eight has a vertical velocity component, so the lumped drag acquires a vertical part that grows with speed (its driver $v_z \|\boldsymbol{v}\|$ rises from 0.02 to $14\,\mathrm{m^2\,s^{-2}}$ in RMS across the battery), it is uncorrelated with the applied steps, and the vertical steps are half the size of the horizontal ones (0.8 against 1.5 N on x; the vector magnitude is the $\sim$2 N of Section VI), so the same uncorrelated content costs more correlation on that axis. The absolute vertical error moves the other way, from 0.62 N in hover, where the mass–force ambiguity biases the plain filter's d_z (a bias the correlation does not see), to 0.29 N at $10.2\,\mathrm{m\,s^{-1}}$; the dip to 0.91 at $6.1\,\mathrm{m\,s^{-1}}$ is comparable to the run-to-run spread ($\pm$0.02). A moving-horizon estimator matches the EKF on these and does not improve them, confirming that the force needs no special estimator beyond the accelerometer residual. Figure 2 shows the estimate tracking the disturbance in the time domain on all three axes.

The reading itself is the translational analogue of a classical momentum/residual observer [7], [8], and we do not claim to improve it: any such observer computed with the same mass would return the same force. The comparison that matters is therefore not against another force estimator but against the same estimator without the calibration layer, which is the ablation of the following subsections and quantifies exactly the mass bias that an uncalibrated, or unconditionally calibrated, observer inherits.

We characterize the sensor with a calibration curve, the standard test for a force sensor: the estimated force ($m\hat{d}$) against the applied force, per axis, over a sinusoidal wind that sweeps the range (Fig. 4). The fit is linear and correctly scaled, with slopes 0.98, 0.97, 0.96 for the three axes, sub-0.2 N intercepts, and R^2 of 0.90, 0.89, 0.87; the unity slope confirms the reading is correctly scaled by the mass and the small residual is the bandwidth limit of the random-walk force model. The sensor is bandwidth-limited rather than range-limited: an amplitude sweep of step disturbances shows the slope holding above 0.8 up to about 2 N and then falling (to 0.4 at 4 N) as large, fast steps exceed the random-walk bandwidth, so the usable full scale is set by how quickly the disturbance varies, a limitation a higher-order force model would relax.

B. MASS IDENTIFICATION AND ITS HOVER LIMIT

Given excitation and no confounding disturbance, the mass is identified accurately. Figure 5 shows the EKF converging from a deliberately wrong prior (0.60 kg, 43 % low) to within 0.6 % of the ground truth on a disturbance-free trajectory, with the estimated force correctly near zero. This is the

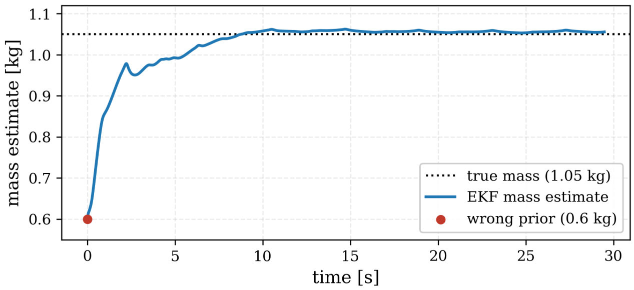

FIGURE 5. **Mass identification under excitation: from a $0.60\,\mathrm{kg}$ prior to within $0.6\,\%$ (disturbance-free trajectory).**

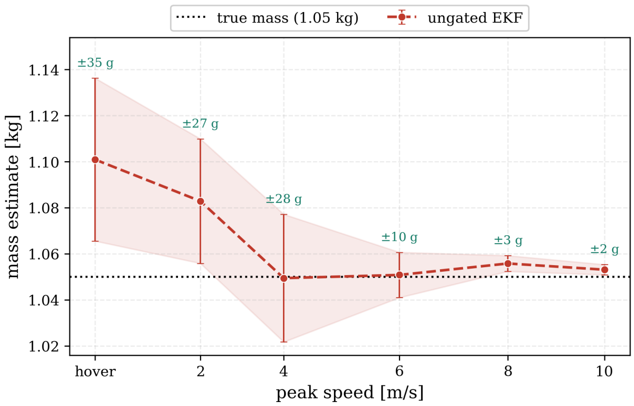

FIGURE 6. **Mass identifiability against excitation, *ungated* EKF (N=10 per speed, N=9 in hover; mean $\pm$ std).**

identification result: the mass is a physical invariant and is recovered to the true value once the trajectory disperses the thrust vector.

Hover is where identification is hard, and it exposes the limit of the ungated filter. Figure 6 shows the plain EKF mass against speed under wind *without* the gate: in hover the estimate is biased by $4.8\,\%$ and varies by $\pm 35\,\mathrm{g}$ across the $N = 9$ runs, because the vertical force is misread as a mass change (Proposition 1); under excitation the bias falls below $1\,\%$ and the spread to $\pm 2\,\mathrm{g}$, in agreement with the σ^{-1} conditioning of Section III. This hover bias, which appears even when the mass is seeded from the correct value, is exactly what the gate prevents, as the next experiment shows.

C. THE MASS GATE PROTECTS THE ESTIMATE

Figure 7 flies an aggressive trajectory followed by a hover, both under wind, over five runs with independent noise and wind. During the trajectory both filters identify the mass. In the hover that follows, the plain EKF is blind to the lost identifiability and attributes the vertical wind to a mass change; its drift is more pronounced here than in the steady-hover battery of Figure 6 ($4.8\,\%$), because the maneuver leaves the filter with an inflated mass covariance, so the wind moves the estimate by $7\,\%$ over the transition (7.0 ± 0.6, N=5). The identifiability-aware filter detects $\tilde{\sigma}$ falling below $\tilde{\sigma}_{\min}$ and gates the mass update, cutting the drift to $3\,\%$ (3.3 ± 0.8);

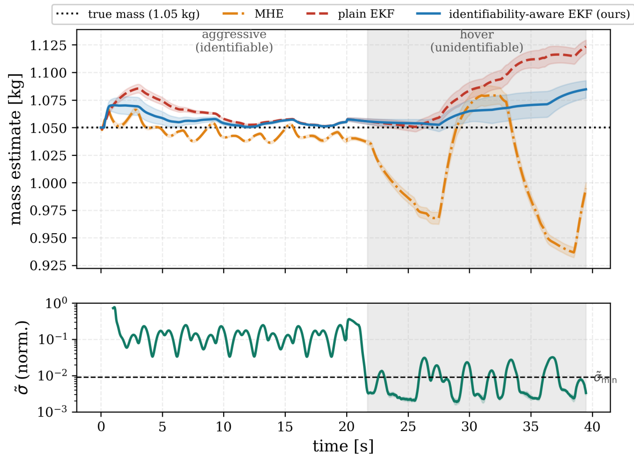

FIGURE 7. **Mass over a windy maneuver-to-hover transition (N=5, mean $\pm$ std); hover shaded, $\tilde{\sigma}$ and threshold at the bottom.**

a residual movement remains because the windy hover still excites $\tilde{\sigma}$ intermittently, but the gate more than halves the error and is consistently better than the plain EKF across all runs. The gate protects an estimate that hover can no longer fully support.

This advantage comes specifically from the Schur complement, not merely from gating on excitation. We ran the same filter with the gate keyed instead to a generic persistency-of-excitation proxy, the total regressor energy $\sum_k \|\mathbf{\Phi}_k\|^2$, a conventional dead-zone. Because hover is high in thrust energy though zero in dispersion, this proxy never closes and the result coincides with the ungated plain EKF: on five repetitions the energy-gated and the plain estimates are identical at the end of the hover ($6.4\,\%$ to $8.7\,\%$ drift on a new set of noise realizations). A generic excitation measure cannot tell hover from a maneuver. Only $\tilde{\sigma}$, which measures the dispersion that isolates the mass–vertical-force coupling, detects the loss of identifiability and protects the calibration.

D. ACTIVE EXCITATION RECOVERS AN UNIDENTIFIABLE MASS

When the mass is unknown and the vehicle must hover, no passive estimator can recover it. Figure 8 starts from a deliberately wrong prior ($0.70\,\mathrm{kg}$) in a wind-free hover, the same conditions for the passive and the active filter. The passive filter stays frozen at the prior, since hover provides no information (the passive estimate stays at $-33.3\,\%$ across all N=5 runs); the active controller injects the probe of (22), $\tilde{\sigma}$ rises above the threshold, the gate opens, and the mass converges from 0.70 toward the ground truth; after $15\,\mathrm{s}$ of gate-open time the probe is withdrawn, the gate closes, and the estimate is held for the remaining $34\,\mathrm{s}$ of hover (the slow rise of $\tilde{\sigma}$ after withdrawal is the noise-driven thrust jitter of the hover and stays a factor of two below the threshold). Using the most informative probe of Section VII-E, the mass error falls from $33\,\%$ to below $1\,\%$ (0.4 ± 0.2, N=5), the same accuracy the filter reaches on the disturbance-free trajectory of Figure 5

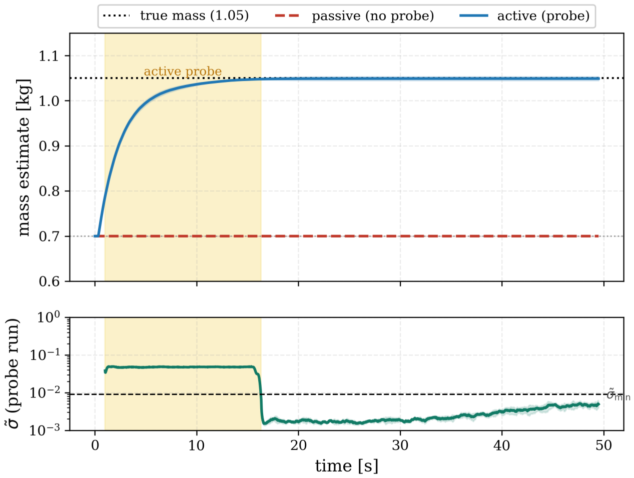

FIGURE 8. Mass recovery from a wrong hover prior (0.70 kg; wind-free hover; N=5, mean ± std): the probe (shaded) lifts σ, recovers the mass, and is withdrawn after 15 s of gate-open time; the estimate is then held for the rest of the hover.

(0.6 %): fifteen seconds of probe bring the estimate to the accuracy the trajectory gives. How fast the recovery proceeds depends on the probe shape, which the next experiment examines.

E. SHAPING THE PROBE BY IDENTIFIABILITY

The speed of the recovery in Fig. 8 depends on the probe shape, and the identifiability tells us which shape is best: since the Schur complement is the Fisher information on the mass (Proposition 2), the probe that raises it most per unit deviation tightens the bound fastest. We sweep the vertical/lateral mix of (22) at the fixed amplitude budget $\rho = 0.8$ m, from a pure lateral tilt to a pure vertical bob, and measure the $\tilde{\sigma}$ achieved while the probe is active and the time for the mass error to settle below 2 % (Fig. 9, N=5 per shape). The achieved $\tilde{\sigma}$ rises monotonically from 0.033 for the lateral tilt to 0.048 for the vertical bob, and the settling time falls in the same order, from 14.3 ± 1.3 to 8.3 ± 0.6 s (one-sided Mann–Whitney against the lateral tilt, $p = 0.004$), consistent with Proposition 2: the information rate sets how quickly the bound tightens. All five shapes exceed the threshold at this budget and all recover the mass to 0.4 % to 1.2 %, since the probe is withdrawn only after 15 s of gate-open time whatever its shape. The ranking is the opposite of the one we obtained with a preliminary 0.4 m budget, where only the lateral tilt exceeded the threshold and the vertical bob raised $\tilde{\sigma}$ least, because the controller attenuates a commanded 1 Hz bob of that amplitude: what counts is the $\tilde{\sigma}$ the closed loop actually achieves, not the one the reference prescribes. This is the practical reason for shaping the probe by the measured $\tilde{\sigma}$ rather than by a fixed rule, and the shape is optimal only within the two-parameter family we sweep, not the unconstrained optimal input.

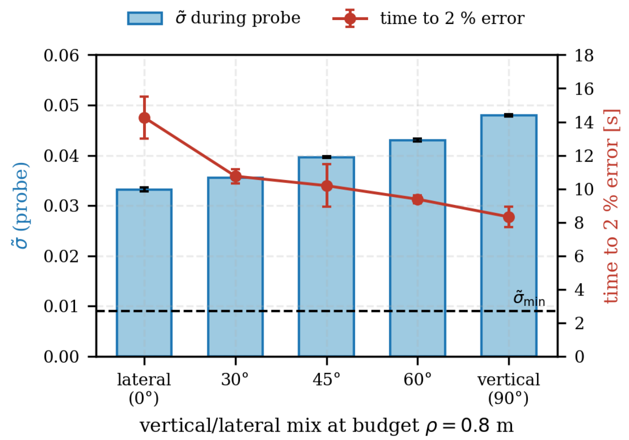

FIGURE 9. Probe shaping at the 0.8 m budget: identifiability achieved while the probe is active (bars) and time for the mass error to settle below 2 % (line), against the vertical/lateral mix (N=5 per shape, mean ± std).

TABLE 5. Disturbance rejection (N = 5 per mode; hover N = 4): tracking RMSE with and without the force feedforward under the same wind, against peak speed.

Peak [m s^{-1}]	No ff [cm]	ff [cm]	Reduction
hover	38.7	8.5	78%
2.1	38.0	7.8	80%
3.8	40.0	8.4	79%
6.1	36.0	8.4	77%
8.2	32.7	14.6	55%
10.2	32.8	21.6	34%

F. DISTURBANCE REJECTION

Feeding the estimated force to the controller reduces the tracking error under wind (Fig. 10), and the reduction scales with how much the wind dominates the dynamics (Fig. 11, Table 5). Up to 6 m s^{-1} the wind dominates and the feedforward cuts the error by 77 to 80 % (78 % in hover); as the speed grows the wind becomes a small fraction of the aerodynamic forces and the reduction falls to 34 % at 10 m s^{-1}. The improvement is statistically significant at every speed: a one-sided Mann–Whitney U test of the feedforward against the baseline rejects the null at each operating point ($p < 0.01$, $n = 5$ per condition; $p = 0.008$ for the $n = 4$ hover case), with a large pooled effect size (Cohen's $d = 5.7$). The estimator and controller each run at 100 Hz with a per-step time below 1 ms on the laptop-class processor of Section V. Figure 10 shows the spatial effect on the figure-eight: with the feedforward the vehicle stays on the reference, while without it the wind pushes it off the path.

G. ROBUSTNESS

The virtual force sensor is not specific to the step disturbance used above. Under a sinusoidal wind, a continuously varying profile, the estimate tracks the ground truth with a per-axis correlation near 0.93 (Fig. 12), slightly below the step value because the moving disturbance has more bandwidth. All results so far use the simulator's injected odometry noise,

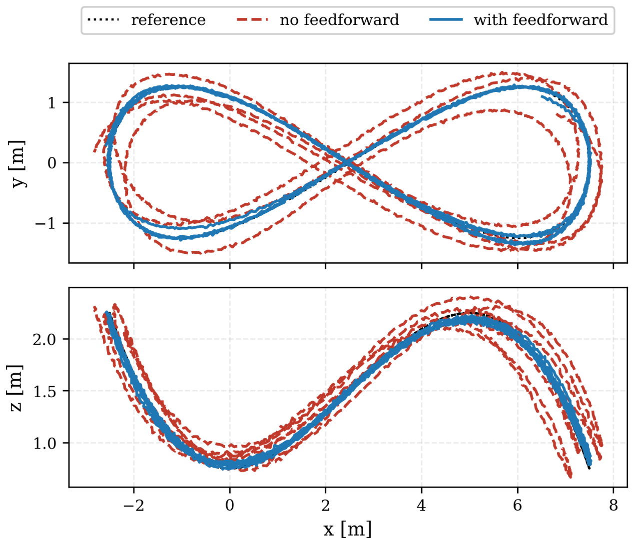

FIGURE 10. Trajectory under wind: with the feedforward (blue) versus without (red); reference dotted.

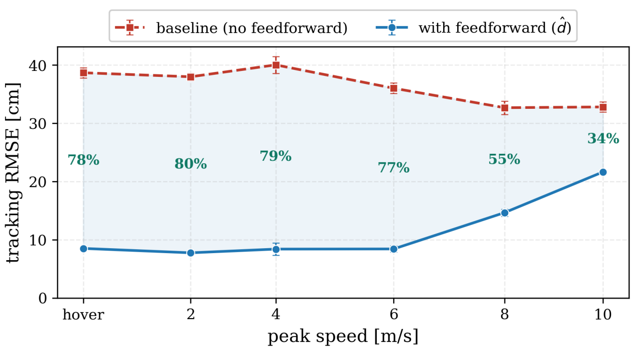

FIGURE 11. Disturbance rejection against speed (N=5 per mode; hover N=4; mean $\pm$ std).

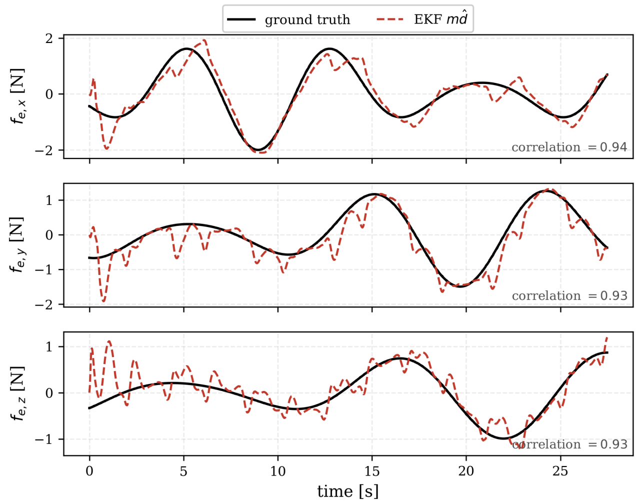

FIGURE 12. Virtual force sensor under a sinusoidal wind (representative run); per-axis correlation near 0.93.

so the estimator is already exercised on noisy measurements rather than a clean ground-truth state. The gate threshold is stable across maneuvers once the identifiability is normalized. The raw σ separates the regimes but its absolute scale follows the thrust level, so it required a maneuver-specific threshold (a span of more than an order of magnitude across our experiments). The dimensionless $\tilde{\sigma}$ of (11) removes that dependence: hover sits at $\tilde{\sigma} \approx 0.002$ (steady hover, still or windy: Fig. 8 and Table 6) to 0.005 (the post-maneuver windy hover of Fig. 7, $1\,\mathrm{s}$ window median) while the fastest trajectory reaches 0.46 (Table 6), two orders of magnitude apart. A single $\tilde{\sigma}_{\min} = 0.009$ classifies hover, maneuver, and the active probe alike: the probe lifts $\tilde{\sigma}$ to 0.033 to 0.048 , four to five times the threshold (Table 6), while hover keeps the gate closed. How far the threshold, the window, the dwell, the process and measurement noise, and the maneuver intensity can move before the gate degrades is quantified in Section VII-I.

H. ROBUSTNESS TO SENSOR ERRORS

A virtual force sensor lives or dies by the error sources a real IMU and thrust map introduce, which the ideal simulator does not exercise. We inject them into the estimator inputs and sweep their magnitude (Fig. 13): an accelerometer bias up to $0.4\,\mathrm{m\,s^{-2}}$, a scale-factor error up to $10\,\%$, and a thrust-coefficient mismatch up to $10\,\%$. The ranges are chosen to bracket what a real platform presents before any in-field calibration: the bias range covers the uncalibrated initial offset of consumer MEMS accelerometers (tens of $\mathrm{m}g$, i.e. a few tenths of $\mathrm{m\,s^{-2}}$), the scale-factor range covers their sensitivity tolerance of a few percent with a wide margin, and the thrust range covers the change of the thrust map over a flight from battery-voltage sag and propeller wear, which is of the order of $10\,\%$. The sensor degrades gracefully and proportionally, with no breakdown. The force error rises from $0.42\,\mathrm{N}$ to at most $0.58\,\mathrm{N}$ (under the largest bias), and the mass calibration from below $1\,\%$ to at most $11.5\,\%$ (under a $10\,\%$ thrust mismatch, which couples almost one-to-one into the mass). The bias loads the force but barely the mass, while the scale and thrust errors load the mass through the $\frac{T}{m}$ term, as expected; the near-linear degradation suggests that a one-time bench calibration of the bias and scale may remove much of it, which we have not tested.

I. FISHER INFORMATION, THRESHOLD, AND GATE SENSITIVITY

Proposition 2 bounds the variance of the mass estimate; we check it against the spread actually observed and then examine how sensitive the gate is to its parameters. Figure 14(a) plots, against speed, the across-run standard deviation of the plain-EKF mass from Table 4 together with the constant-force bound of Proposition 2 and the random-walk bound of (15), both computed from the logged thrust and attitude with the filter's r_s and q_d and accumulated over the full run

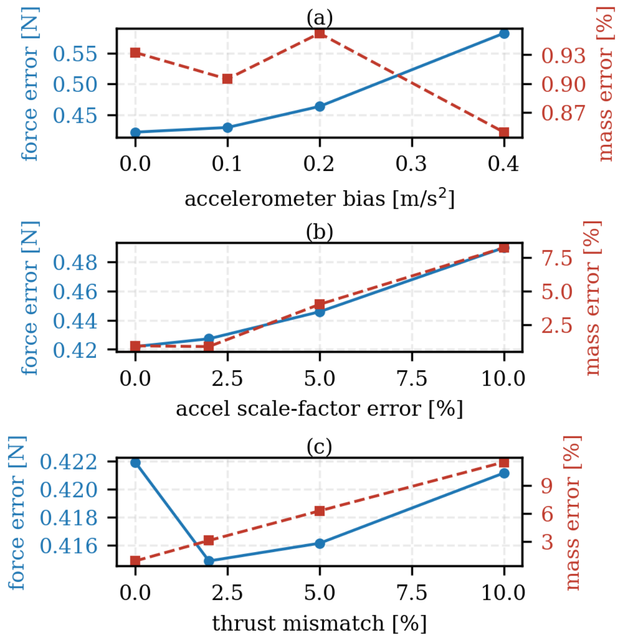

FIGURE 13. Robustness to injected accelerometer bias, scale-factor error, and thrust mismatch: force error (blue, left axis) and mass error (red, right axis).

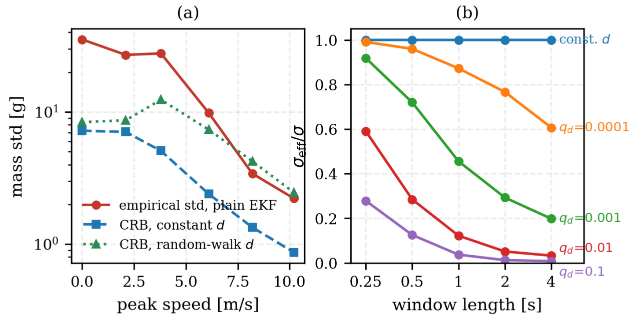

FIGURE 14. Fisher information on the mass from logged flight data. (a) Across-run standard deviation of the plain-EKF mass against speed, with the constant-force bound and the random-walk bound. (b) Ratio of the random-walk to the constant-force information against window length ($0.25\,\mathrm{s}$ to $4\,\mathrm{s}$) for q_d from 0 to 0.1 per step.

TABLE 6. Classification signal against information content, per regime, over $1\,\mathrm{s}$ windows (median over the runs of Table 4; probe: gate-open windows of Fig. 8). Last column: Cramér–Rao lower bound on the mass standard deviation from one window (Proposition 2), the best resolution the window could give.

Regime	$\tilde{\sigma}$	σ/r_s	CRB std($\hat{m}$) [g]
hover	0.0024	47	160
$2.1\,\mathrm{m\,s^{-1}}$	0.0023	48	160
$3.8\,\mathrm{m\,s^{-1}}$	0.014	298	64
$6.1\,\mathrm{m\,s^{-1}}$	0.12	3413	19
$8.2\,\mathrm{m\,s^{-1}}$	0.30	17330	8.4
$10.2\,\mathrm{m\,s^{-1}}$	0.46	48739	5.0
probe (vertical bob)	0.048	1068	34

(unlike the per-window values of Table 6); the random-walk information over a $1\,\mathrm{s}$ window is $12\,\%$ of the constant-force value at the implemented q_d (Fig. 14(b)). Under excitation ($6.1\,\mathrm{m\,s^{-1}}$ to $10.2\,\mathrm{m\,s^{-1}}$) the empirical spread falls from 10 to $2.2\,\mathrm{g}$ while the constant-force bound falls from 2.4 to $0.9\,\mathrm{g}$; scaling by the random-walk ratio of Fig. 14(b) brings the bound to $2.5\,\mathrm{g}$ to $7\,\mathrm{g}$, the scale of the spread (it can even exceed it at the two fastest speeds, since the across-run spread of a recursive estimate that accumulates many windows is not the per-window variance the bound addresses), so the Fisher information of Section III describes the estimator that is actually implemented once the force model is accounted for. Near hover the spread ($35\,\mathrm{g}$) exceeds both bounds by a wide margin because the error there is a *bias*, the vertical wind misread as mass, which no variance bound captures: this is Proposition 1 seen from the estimator side and the reason the gate is needed.

Table 6 separates the two roles of the identifiability signal. The classification signal $\tilde{\sigma}$ spans two orders of magnitude from hover to the fastest trajectory, and the probe sits well above the threshold, as intended. The information content σ/r_s spans three orders of magnitude over the same regimes because it retains the thrust level; the last column converts it through Proposition 2 into the lower bound on the mass standard deviation that one second of data could achieve. The probe, at $\tilde{\sigma} = 0.048$, carries per second the information of a flight between 3.8 and $6.1\,\mathrm{m\,s^{-1}}$; because all the near-hover regimes share the same thrust level, $\tilde{\sigma}$ and σ/r_s rank them identically here, but the table makes explicit that equal $\tilde{\sigma}$ at different thrust or noise levels would not mean equal information.

The window length and the threshold are coupled, and we quantify the coupling offline on the transition runs of Fig. 7 by recomputing $\tilde{\sigma}$ from the logged thrust and attitude for windows of $0.25\,\mathrm{s}$ to $4\,\mathrm{s}$. The ratio between the maneuver and the hover medians of $\tilde{\sigma}$ is 5 at $0.25\,\mathrm{s}$, 15 at $0.5\,\mathrm{s}$, peaks at 28 at the $1\,\mathrm{s}$ window used, and falls to 16 and 9 at 2 and $4\,\mathrm{s}$, because a short window sees too little of the maneuver and a long one lets the windy hover accumulate dispersion (its median $\tilde{\sigma}$ rises from 0.0025 at $0.25\,\mathrm{s}$ to 0.033 at $4\,\mathrm{s}$). The threshold of 0.009 therefore sits between the two medians, with a margin of about two on the hover side and more than ten on the maneuver side for windows of $0.5\,\mathrm{s}$ to $1\,\mathrm{s}$, and would have to be raised for windows of $2\,\mathrm{s}$ or longer. Within any window the windy hover still produces transient spikes (its 95th percentile at $1\,\mathrm{s}$ is 0.027), which is what the dwell rejects.

Table 7 reports the sensitivity study proper. Each row repeats the windy maneuver-to-hover transition of Fig. 7 with one parameter changed at a time, five seeds per setting, and reports the root-mean-square mass error over the settled hover for the gated and the plain filter; the reference setting of Table 3 gives 2.2 ± 1.0 against $5.1 \pm 1.1\,\%$ (an RMS over the hover, hence lower than the end-of-transition values of 3.3 and $7.0\,\%$ of Fig. 7). *Threshold*: below 0.009 the gate re-opens on the windy-hover excursions and its advantage shrinks ($3.4\,\%$ to $3.8\,\%$); at 0.015, which sits at the level of those excursions, it opens intermittently with the frozen covariance and becomes erratic (5.1 ± 2.8 %); at 0.03 to 0.06 it stays shut through the whole hover and holds the mass to 0.4 % to 0.6 %. The threshold is therefore limited not by the hover but by what must re-open the gate: the probe and the slowest flight that should still identify the mass. With the 0.80 m budget the probe reaches $\tilde{\sigma} \approx 0.048$ (Table 6; a preliminary 0.40 m budget left it at the threshold, Section VII-E); with it a threshold of 0.03 would be admissible for the probe and would hold the mass below 1 %, but it would exclude the 3.8 m s^{-1} flight ($\tilde{\sigma} = 0.014$) from identification, so we retain 0.009 and accept the residual re-openings in windy hover. *Dwell*: the gate is never worse than the plain filter for dwells of 0.1 s to 1 s and is best at the 0.25 s used; a short dwell lets gusts open it too often, and a long one makes re-openings rare but larger, since each is made with the covariance frozen since the maneuver (the end-of-hover value at 1 s, not listed in Table 7, scatters by ±7 %). *Window*: 0.5 s is too short to separate the regimes and the gate degrades to 8.8 %; 1 s to 2 s hold at 2.2 % to 3.1 %. *Process noise*: the gate helps for $q_m \leq 10^{-5}$ and is counterproductive at $q_m = 10^{-4}$, where a mass that adapts within a second no longer integrates the hover bias and the plain filter is already at 3.6 %. The force process noise sets how much of the hover ambiguity the filter resolves by fiat: at $q_d = 10^{-3}$ the plain filter attributes the wind almost entirely to the mass and drifts by 53 %, which the gate cuts to 6.9 %; at $q_d = 10^{-1}$ the force absorbs everything, the plain filter drifts by only 1.8 %, and the gate is unnecessary (4.7 ± 2.6 %). A loose force model hides the ambiguity rather than resolving it, and is correct only if the mass never changes; the gate makes the same decision explicit and conditional on identifiability, and leaves q_d free to be set for the force reading. *Accelerometer noise*: added white noise of 0.3 m s^{-2} to 0.7 m s^{-2} leaves the gated error at 2.2 % to 2.5 %; the plain filter's own drift shrinks (1.9 % to 2.9 %) because a noisier residual moves the mass less, so the margin narrows. *Maneuver intensity*: at peak speeds of 5.9 m s^{-1} to 8.4 m s^{-1} the picture is unchanged (2.2 to 2.7 against 5 %); at 3 m s^{-1} the maneuver never lifts $\tilde{\sigma}$ above the threshold, neither filter identifies the mass, the mass covariance never contracts, and both are vulnerable to the transient at the hover onset (44 % plain, 40 ± 44 % gated, one run in five latching a wrong value after a brief opening). This is the regime the passive gate cannot handle by construction, since it protects an estimate that was never formed, and the one the active probe of Section V-B exists for. The operating region of the gate is therefore a window of 1 s to 2 s, a dwell of about 0.25 s (longer dwells keep the RMS error but make the end value erratic), $q_m \leq 10^{-5}$, $q_d \leq 10^{-2}$, and a threshold between the hover excursions and the probe level, with the probe budget as the lever that widens it. Transfer to other vehicles, thrust-to-weight ratios, and IMU grades is not established by one simulated platform; (16) provides a necessary best-case information condition, not a sufficient accuracy guarantee, for selecting a threshold from N, r_s, and the desired mass uncertainty.

TABLE 7. Gate sensitivity on the windy maneuver-to-hover transition of Fig. 7: root-mean-square mass error over the hover [%], mean ± std over five seeds, for the gated and the plain filter. One parameter is changed at a time from the reference setting of Table 3.

Parameter	Value	gated	plain
threshold $\tilde{\sigma}_{\min}$	0.003	3.4 ± 0.3	4.6 ± 0.7
	0.006	3.8 ± 0.9	4.6 ± 0.3
	0.009 (ref.)	2.2 ± 1.0	5.1 ± 1.1
	0.015	5.1 ± 2.8	4.8 ± 0.2
	0.03	0.4 ± 0.3	4.5 ± 0.4
	0.06	0.6 ± 0.2	4.4 ± 0.5
dwell t_d [s]	0.1	3.5 ± 0.9	4.8 ± 0.3
	0.25 (ref.)	2.2 ± 1.0	5.1 ± 1.1
	0.5	4.1 ± 1.4	4.4 ± 0.8
	1	3.1 ± 1.6	4.7 ± 0.6
window [s]	0.5	8.8 ± 3.9	4.7 ± 0.5
	1 (ref.)	2.2 ± 1.0	5.1 ± 1.1
	2	3.1 ± 0.6	4.7 ± 0.4
q_m	10^{-6}	0.7 ± 0.3	1.8 ± 0.4
	10^{-5} (ref.)	2.2 ± 1.0	5.1 ± 1.1
	10^{-4}	7.1 ± 2.4	3.6 ± 0.5
q_d	10^{-3}	6.9 ± 0.4	52.7 ± 0.1
	10^{-2} (ref.)	2.2 ± 1.0	5.1 ± 1.1
	10^{-1}	4.7 ± 2.6	1.8 ± 0.7
accel. noise [m s^{-2}]	0 (ref.)	2.2 ± 1.0	5.1 ± 1.1
	0.3	2.5 ± 1.1	1.9 ± 0.9
	0.7	2.2 ± 0.7	2.9 ± 0.7
peak speed [m s^{-1}]	3.0	39.6 ± 43.6	44.4 ± 2.3
	5.9 (ref.)	2.2 ± 1.0	5.1 ± 1.1
	8.4	2.7 ± 1.2	5.0 ± 0.4

TABLE 8. Estimator comparison on identical data. Force entries are pooled over the battery; the excited-mass entries pool the well-excited runs (8.2–10.2 m s^{-1}, N=10 each speed); hover and recovery entries are from Figs. 7 and 8.

	MHE	plain EKF (baseline)	gated EKF (ours)
horizontal force corr	0.96	0.97	0.97
vertical force corr	0.66	0.95	0.95
mass error, excited	+0.4%	+0.4%	+0.4%
mass error, windy hover	5.3%	7.0%	**3.3%**
recover mass, wrong prior	no	no	yes

J. ESTIMATOR COMPARISON

Table 8 consolidates the comparison on the identical data. The two baselines fail in different ways, both traceable to the hover ambiguity of Proposition 1. The plain EKF reads the force well, including the vertical axis, but lets the hover mass drift as the vertical wind is misread as a mass change. The moving-horizon estimator is worse: its free mass both corrupts the vertical force, which it misreads as mass, and drifts in hover (Fig. 7), and the window adds cost without recovering either. Our identifiability-aware EKF keeps both: the EKF structure holds the vertical force, and the gate keeps the mass closest to the truth in the windy hover (3.3 %, against 7.0 % for the plain EKF and 5.3 % for the MHE). With active excitation it is also the only one that recovers the mass from a wrong prior in hover. All three run in real time at 100 Hz.

VIII. CONCLUSION

We built a self-calibrating virtual force sensor for quadrotors. The external specific force is read directly from the accelerometer; its only calibration parameter, the mass, is identifiable only under excitation and is confounded with a vertical force in hover. We made the calibration identifiability-aware on two levels: an extended Kalman filter that gates the mass calibration by the online identifiability $\tilde{\sigma}$, freezing it when it cannot be seen, and a model-predictive controller that injects a minimal probe to restore identifiability when the mass must be recalibrated but the flight lacks the excitation. In software-in-the-loop the sensor reads the external force to about $0.4\,\mathrm{N}$ RMS per axis (per-axis correlation 0.93 under continuous wind), the gate holds the mass across a windy hover transition to $3\,\%$ where a plain EKF drifts to $7\,\%$, the active probe recovers the mass from a wrong prior that hover alone cannot identify, and the force feedforward reduces the tracking error by 34 to $80\,\%$ under wind, scaling with how much the wind dominates the dynamics.

A matched comparison showed that a moving-horizon estimator does not improve on the EKF for this problem, because the informative measurement is instantaneous; the contribution therefore lies in the identifiability logic, not in the estimator class. The dead-zone and the excitation are each classical; what ties them together is the closed-form Schur complement, the Fisher information on the mass computed from known signals (its inverse is the Cramér–Rao bound), whose dimensionless form $\tilde{\sigma}$ drives the gate and the probe from one online scalar, including the choice of the most informative excitation. The sensitivity study bounds where this holds: the gate helps for windows of $1\,\mathrm{s}$ to $2\,\mathrm{s}$, a dwell of about $0.25\,\mathrm{s}$, a mass process noise at or below 10^{-5}, and a force model stiff enough to read the force ($q_d \leq 10^{-2}$), and it cannot help where the flight never identifies the mass, which is the regime the probe exists for; the probe, in turn, is shaped by the identifiability the closed loop actually achieves, which is why it is measured rather than prescribed. The main limitation is that the study is in simulation; hardware validation with a motion-capture system and a calibrated disturbance is the immediate next step, as is a continuous (rather than family-restricted) optimal-input design. Because the filter reads any external specific force, the approach extends toward contact and tether forces, where an identifiability-aware wrench estimate would support aerial physical interaction [34], [35] without a dedicated force sensor.

CODE AVAILABILITY

The source code and experiment scripts are openly available at https://github.com/bryansgue/observability-aware.

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